color, but a spread across frequencies—a spectrum. Characteristic patterns in the spectrum of a star are due to bits of the light being absorbed or emitted by different chemical elements in the star's atmosphere. When you spread out the light through a prism, different colors appear at different intensities, and dark lines or gaps show up at those specific frequencies where atoms in the star's atmosphere have absorbed the light—the light at those frequencies was removed by the gas before it could reach you. These features produce a kind of bar code unique to each element, with a pattern of lines that astronomers can recognize at a glance. So, for instance, light passing through a cloud of hydrogen will appear with a specific comblike pattern of dark lines when it's spread out across all the frequencies. We know from laboratory tests where the lines should be, and what the pattern should look like, and we can repeat the process with the patterns from other elements as well. If a star has a recognizable comblike pattern in its spectrum, but the lines appear at the "wrong" frequencies, that indicates the light from the star has been shifted by the star's motion. If each line is shifted in the same way to lower frequencies, that's a redshift and the star is moving away. If each line is shifted high, that's a blueshift and the star is approaching. And how far the lines have shifted tells us how quickly the star is moving.

Astronomers have gotten very good at this kind of measurement. Redshift or blueshift is now one of the most straightforward things to observe about any source of light in the universe, provided a spectrum is taken and it has any recognizable line patterns at all. We can use it to see how stars within our galaxy move relative to us, or to detect the tiny wobble of a star being pulled back and forth by the orbit of a planet around it.

And when it comes to distant galaxies, we can now use the redshift to measure not only how they're moving relative to us—toward us or away, and how quickly or slowly—but how far from us they are while they're doing it. How does that work? The expansion of the universe means that however a galaxy might be moving through its own space, the fact that the space between here and there is expanding means that it will also, in general, be moving away from us. And how quickly it's moving away from us depends on how far away it is now.

In 1929, astronomer Edwin Hubble was looking at galaxy redshifts when he noticed a striking, conveniently simple pattern. Galaxies that are farther away have, on average, higher redshifts. This relationship has allowed us to